

Nafis Anwar

Software Engineer

656-203-8046 | anwarnafisneo@gmail.com | [linkedin.com/in/nafisanwar/](https://www.linkedin.com/in/nafisanwar/) | nafisanwar.github.io | Tampa, FL

Summary

Software Engineer with strong C++/Java/Python foundations in data structures, algorithms, AI and systems. Built backend and tooling projects including a Dockerized RSA platform with REST APIs, a WebAssembly C++ VM + React IDE, and a FAISS semantic search system over 28k+ posts.

Education

University of South Florida

Bachelor of Science in Computer Science

Tampa, FL

Expected Graduation: May 2027

Relevant Coursework: Data Structures, Analysis of Algorithms, Object-Oriented Programming, Software System Development, Computer Organization, Computer Architecture, Machine Learning, Artificial Intelligence (Search & Planning), Reinforcement Learning, Robotics Simulation & Control, Probability & Statistics

Experience

Junior Undergraduate Researcher

RARE Lab (USF)

Dec 2025 - Mar 2026

Tampa, FL

- Operate and maintain the **Unitree G1 humanoid robot**, configuring power-on sequences, scripts, evaluation harnesses (logging, configuration management, benchmarking, regression checks) to support research experiments and ensure safe, reproducible robot operations.
- Conducted **literature reviews on sim-to-real transfer techniques** including **physics mismatch mitigation**, **reward shaping strategies**, and **delta-action policies**, compiling findings into technical summaries to further team knowledge and inform experimental design.
- Analyzed research codebases (**ASAP framework**) to reproduce published results, trace training pipelines, and identify core policy learning and evaluation modules, establishing benchmarks for humanoid locomotion experiments.

Resident Assistant

Residential Education at USF

Aug 2025 - Present

Tampa, FL

- Built **automated reporting/tracking** for community metrics using **Excel macros** and **ASP.NET Core**; documented 18 conflict-resolution cases and 58 issue reports while supporting **500+ residents**, which streamlined reporting and improved response efficiency
- Fostered community engagement by organizing and facilitating **4 social and educational programs**, averaging 20 attendees, and boosted floor participation by **36% for 34 residents** within my community.

Projects

Stock Analysis Visualizer | C#, .NET WinForms

Oct 2025 - Nov 2025

- Built a **Windows Forms application to visualize stock OHLC candlesticks** and volume from **Yahoo Finance CSVs** using the .NET charting library, with data binding directly to chart series.
- Added multi-CSV support (day/week/month), normalized OHLC ranges with padding, and indexed trading days to hide weekend/holiday gaps
- Implemented a refresh pipeline that updates charts without reloading files to improve responsiveness
- Wrote **parsing and validation checks** for malformed CSV rows and missing values to reduce crashes and improve reliability

GritVM Web IDE | C++, WebAssembly (Emscripten), React.js, TypeScript, Monaco Editor

Aug 2025 - Sep 2025

- Compiled a **C++ virtual machine to WebAssembly with Emscripten**, exposing functions like load, reset, single-step, run, and state inspection to JavaScript bindings.
- Built a **React + Monaco-Editor IDE** that highlights the current instruction and live-updates register and memory panes on each VM step, using a compact JSON state representation from the WASM module.
- Packaged the **frontend + WASM bundle** and deployed to **Vercel** with **optimized asset caching** so the IDE runs entirely client-side with no backend dependencies.
- Documented VM semantics, debugging workflow, and extension points to help contributors add instructions safely

RSA Full-Stack Educational Playground | C, C#, ASP.NET Core, REST APIs, Docker

Jul 2025 - Aug 2025

- Implemented key-generation, encryption, and signing flows using big-integer operations and modular arithmetic, and profiled different exponentiation strategies to **achieve 30% faster RSA operations than a baseline C# implementation**.
- Wrapped the native library in an **ASP.NET Core** backend via **C# P/Invoke** and exposed **REST endpoints** for key ops with structured JSON requests/responses and robust error handling.
- Containerized the service with **Docker** (build toolchain + native lib + runtime) and documented example curl/client calls so students can use the API consistently across environments.

MediTrends | Python, Flask, FAISS, TensorFlow, PyTorch, Tailwind CSS

May 2025 - Jun 2025

- Built **FAISS-powered semantic search over 28k+ posts** with efficient batching (110 ms/query) and aggregation logic for volume, growth, and sentiment KPIs
- Implemented NLP pipelines for clustering and trend detection; generated KPI-based reports for community-level insights
- Developed **Flask API endpoints** for search and trending and a responsive Tailwind UI for analytical workflows
- Validated retrieval quality by testing **query sets, edge cases, and performance regressions**; documented evaluation notes and tuning decisions
- Designed the system to handle varied data sources cleanly by separating ingestion, embedding, indexing, and API layers
- Added Selenium regression tests for the Flask + UI integration to catch breaking changes in search/trending pages after backend updates

PATHFINDER Obstacle-Avoiding Rover | C, Assembly, Arduino IDE, Embedded Systems, Git

Aug 2023 - Dec 2023

- Led a **cross-functional team of 4** to design and build an obstacle-avoiding rover; **wrote and tested C and Assembly control logic in Arduino IDE** to synchronize ultrasonic sensing with servo motor motion, **improving real-time detection/avoidance by 60%**.
- Debugged timing and sensor-read issues using terminal tools, Serial Monitor, and a logic analyzer, **reducing collisions by 95%** through tighter control-loop behavior and reliability fixes.
- Established a **Git-based workflow** (code reviews + issue tracking) and authored **build/test documentation** so new contributors could reproduce results quickly and extend the system safely.

Technical Skills

Languages: Java, C++, Python, C#, JavaScript, SQL, C, HTML/CSS, Assembly

Backend / APIs: ASP.NET Core, Flask, REST APIs, JSON, P/Invoke

Testing/Automation: Unit testing, integration testing, API testing (Postman), test design, debugging, root-cause analysis, Selenium (UI automation)

Cloud / DevOps: Docker, AWS, Azure, Vercel, Amazon RDS, CI/CD (GitHub Actions)

Data/AI: FAISS, TensorFlow, PyTorch, NLP, clustering, model evaluation, statistics/ML foundations

Tools: Git, Linux/CLI, Arduino